The Cairn Hiking Event Management System (CAR-HMS)

Initial requirements document

Document id: CAR-Req-01

1 Introduction

The initial requirements document captures the purpose of the system from the perspective of the client. In this document, background information on the system and its domain is given as well as the environment and the context of the application.

1.1 Purpose of document

This document identifies the goals and scope of a Hiking Event Management System to be built for the Cairn Hiking Club. It describes the users of the system, the customer who initiated the project, and discusses the functionality that the system must provide to its users.

1.2 Background

The Cairn Hiking Club is a non-profit organization based in the National Capital Region (Ottawa - Gatineau) that organizes hiking events in this region for its members. The club is currently managed by a volunteer board of directors and has a small number of employees to support the organization. The head of the club is Ms. Ava Kami, who used to be a professional mountain guide. Having used many existing online event management systems before, she is not satisfied with the available applications and would like to have a system built in-house that specifically suits the needs of the club.

The National Capital Commission (NCC, https://ncc-ccn.gc.ca/) is the federal organization that plans and manages the public lands of the region, including Gatineau Park, the Greenbelt, urban parks, and the Capital Pathway. To ensure safety, the club has always organized its hikes on the NCC's paths and trails, and relies on the trail information the NCC publishes (https://ncc-hub-ncc-ccn.hub.arcgis.com/datasets/c0b7abc0c9a74a0285807ea257ae4e00_1/about) to choose trails, check what is permitted on them, and tell members where to meet.

The club is particularly interested in an event management system offering more control over how events are matched to members, so that they can find events suitable for their skill level and schedule. This document was developed as part of the first meeting with the club to capture the requirements for a new application that would replace the tools the club currently relies on.

The club's website and membership list are currently maintained by Rio Valden, a board member who works as a software developer at an Ottawa consulting firm. His employer supports the club as a community sponsor and has agreed that Rio and two colleagues will maintain the new system after its first release, on volunteer time. The firm builds its own products with Angular and TypeScript for the frontend and Kotlin with Spring Boot for the backend, and it will only take over a system written in that tech stack.

1.3 Project scope

The project consists of the development of a web-based application that will be used for managing hiking events for the club, and sharing information about the events with the club members and the general public. The club only organizes hikes on the NCC's paths and trails and does not maintain trail information itself. The physical event organization, including the actual hiking activities, is not part of the system's scope. Additionally, the human resource management of the club, including membership management and the new member invitation process, is not part of the system's scope. The system will be used by the club members, the general public, and the club administrators.

1.4 Environment

As a minimum viable product (MVP), users will access the application hosted locally on their own computers, using the latest versions of web browsers. Data will be stored in databases offering security guarantees (i.e. authentication, encryption).

1.5 Glossary

Visitor A person who uses the system without an account. A visitor can browse open hikes. A visitor who is a member of the club can register to become a hiker, using an invitation code received from the club administrators. A visitor cannot sign up for a hike.

Hiker A club member who takes part in hikes. Each hiker is uniquely identified by an email address and registers with an invitation code, a code that the club administrators give to a member when they join the club. Additional hiker information includes name(s), phone number, home address, date of birth, an emergency contact (name and phone), a skill level (such as beginner, intermediate, advanced, expert). A membership has a status (active, expired, suspended). A hiker may only sign up for hikes rated at their skill level or one level above it when their membership is active.

Hiking organizer A hiker approved by an administrator to lead hikes. An organizer holds a leader certification from the club, including certificate number, issue date, and expiry date, and has an organizer level, the highest difficulty they may lead. A hike cannot be created if the organizer's certification expires before the hike date.

Trail A named route that hikes follow, taken from the trail information published by the NCC (see Background). The NCC describes each trail by its name, its type (trail, multi-use pathway, ...), its surface (gravel, soil, stonedust, asphalt, ...), its difficulty (easy, moderate, difficult, most difficult), its open season (year-round, summer, winter), the sector and area it belongs to (Gatineau Park, the Greenbelt, the urban lands), whether parking is available, and its route and location on a map. The club only uses trails on which hiking is permitted.

Hike A scheduled event where a group led by an organizer walks a trail. A hike has a unique identifier, a title, a description, a trail, start date and time, an expected duration, a registration deadline, and a capacity (maximum number of participants). The group meets at the starting point of the trail. The difficulty of a hike is the difficulty of its trail as published by the NCC.

A hike is open as soon as it is created, becomes full when the last spot is taken, is closed when its registration deadline passes, and is considered completed once its date has passed without being cancelled. A hike can be cancelled at any point before it starts. The club records who performed each of these steps. What happens on the day of the hike is not tracked by the system. A hike cannot be scheduled outside the open season of its trail. The registration deadline cannot be more than 2 weeks before the hike date.

Sign-up A hiker's registration for a hike. A sign-up records the time it was made, its status (confirmed, waitlisted, cancelled). When a hike is full, further sign-ups are waitlisted in the order they were made; when a confirmed hiker cancels, the first waitlisted hiker takes the spot.

Administrator An employee or board member of the club who administers the system. Administrators maintain hiker accounts, approve organizers, cancel hikes when needed, and produce reports.

2 Documented requirements

2.1 Functional requirements

- The CAR-HMS shall allow authorized users to create hikes on the NCC's trails.
- The CAR-HMS shall allow hikers to sign up for open hikes and shall keep a waitlist when a hike is full.
- The CAR-HMS shall record the status of each step of a hike's lifecycle: open, full, closed, cancelled.
- The CAR-HMS shall record the user who performed each step of a hike's lifecycle.
- The CAR-HMS shall send hiking reminders before the hike date.
- The CAR-HMS shall enable report generation on trail usage.

2.2 Quality attributes

- All access to the CAR-HMS functionalities other than browsing open hikes shall require authentication and authorization.
- All hiker information shall be stored securely to ensure confidentiality.
- The unavailability of the external service(s) shall not prevent the system from functioning property.

2.3 Technical constraints

Since the system will be maintained by Rio Valden's team after its first release (see Background), the club requires:

- The front end of the system must be implemented in Angular with TypeScript.
- The back end of the system must be implemented in Kotlin with Spring Boot.
- The trail information must be obtained from the NCC's published data.

3 Use cases

3.1 Use case diagram

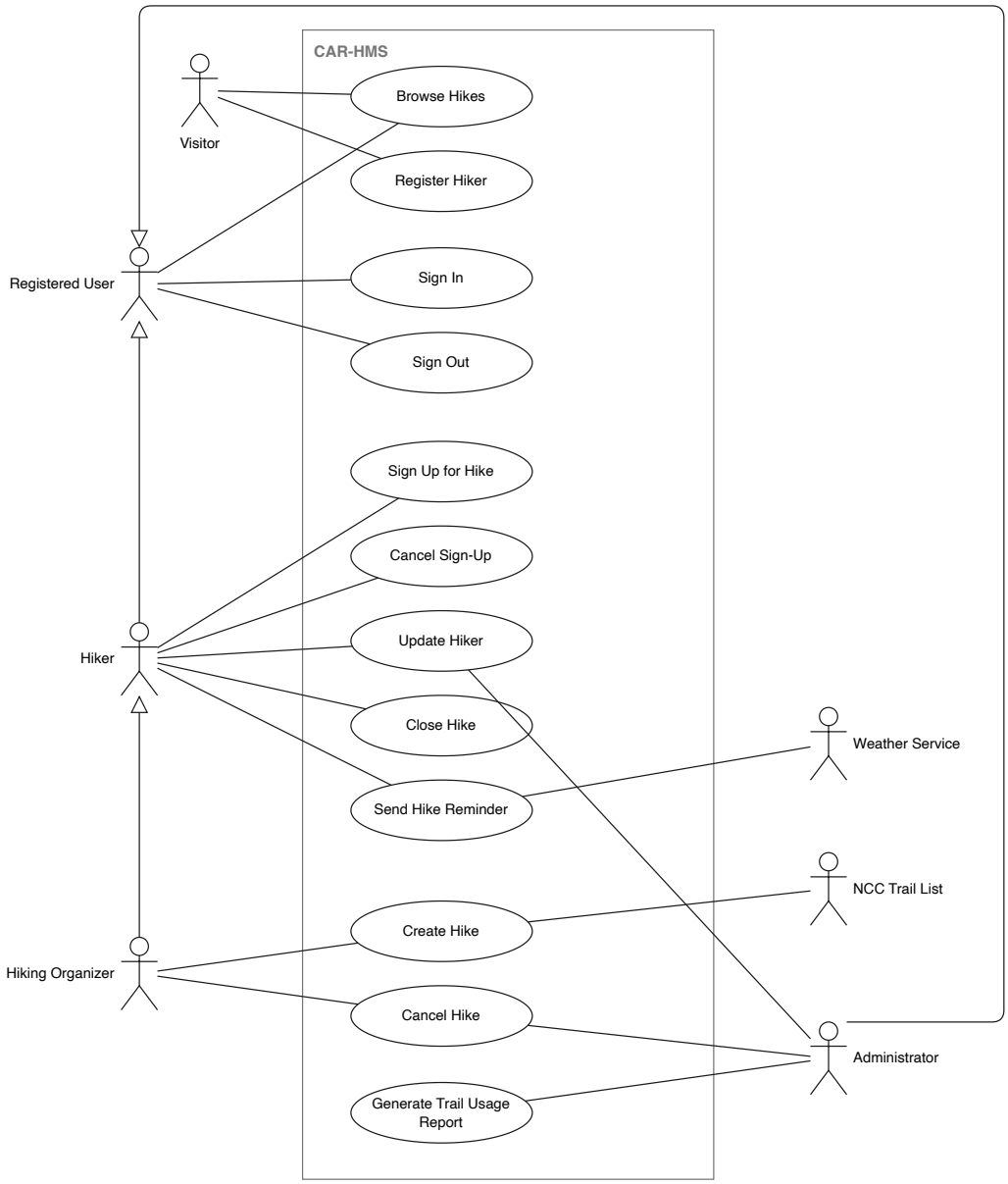


Figure 1: Use case diagram of the CAR-HMS

3.2 Use case summaries

Use case	Priority	Summary
Browse Hikes	5	The Browse Hikes use case allows a visitor or a registered user to view the open hikes. The user may search for a hike by trail name, date, or difficulty, or list all open hikes sorted by date. For each hike the system shows the hike information, the organizer, and the number of remaining spots. A signed-in hiker also sees the hikes matching their skill level.

Use case	Priority	Summary
Register Hiker	6	The Register Hiker use case allows a club member to register as a hiker. The member enters an invitation code received from the club administrators and the hiker information, and accepts a liability waiver. If the hiker is under 18, the waiver must be accepted by a legal guardian, whose name is recorded. Once valid information is entered, the system adds the new hiker with an active membership and the hiker can sign up for hikes.
Sign In	10	The Sign In use case allows a user to identify themselves to the system. The user provides their email and password and, if successfully authenticated, is allowed to use the functionalities of the system in accordance with their role.
Sign Out	11	The Sign Out use case allows a user to log out of the system.
Sign Up for Hike	3	The Sign Up for Hike use case describes how a hiker joins a hike. A signed-in hiker with an active membership selects an open hike (see the Browse Hikes use case) whose difficulty is at most one level above their skill level and confirms. If a spot is available the sign-up is confirmed, and the hike becomes full when the last spot is taken. If the hike is already full, the sign-up is waitlisted in order of sign-up. The system emails the hiker the confirmation or waitlist position.
Cancel Sign-Up	4	The Cancel Sign-Up use case allows a hiker to withdraw from a hike before its registration deadline. The hiker selects one of their confirmed or waitlisted sign-ups and confirms the cancellation. If a confirmed spot opens on a full hike, the first waitlisted hiker is signed up automatically and notified by email. The hike returns to open if spots remain.
Update Hiker	7	The Update Hiker use case allows a hiker to update their information, and allows an administrator to update a hiker’s membership status or to record a leader certification (certificate number, issue and expiry dates) and an organizer level, which makes the hiker a hiking organizer. The system presents a form with the current information that the user modifies and submits.
Close Hike	2	The Close Hike use case describes what occurs when a hike’s registration deadline is reached. The system sets the hike status to closed, after which no further sign-ups or cancellations are accepted.
Send Hike Reminder	9	The Send Hike Reminder use case describes what occurs one day before a hike takes place. The system sends an email to every hiker with a confirmed sign-up, reminding them of the hike’s date, start time, trail, and starting point. The system obtains the weather forecast at the trail’s starting point for the hike date. If rain or snow is predicted,

Use case	Priority	Summary
		the email includes a weather warning. If the forecast cannot be obtained, the reminder is sent without the warning.
Create Hike	1	The Create Hike use case allows a hiking organizer to record a new hike. The trail must be an actual trail in the National Capital Region on which hiking is permitted, according to the trail information published by the NCC (see Background). The hike date must fall within the trail's open season, the registration deadline must be no more than two weeks before the hike date, and the organizer's certification must be valid on the hiking date, with an organizer level at or above the difficulty of the trail. The organizer enters the hike information. If the information is valid and complete, the system assigns a unique identifier to the hike and records it as open. Other hikers can sign up to the hike from that moment.
Cancel Hike	8	The Cancel Hike use case allows the organizer of a hike, or an administrator, to cancel a hike before it starts, giving a reason. The system sets the hike status to cancelled, sets every confirmed and waitlisted sign-ups to cancelled, and notifies those hikers by email with the reason.
Generate Trail Usage Report	12	The Generate Trail Usage Report use case allows an administrator to produce a report on the usage of a trail. The administrator specifies a trail and a period (start date, end date). The report presents statistics on the hikes on that trail in the indicated period (number of hikes, number of participants, number of cancelled hikes).